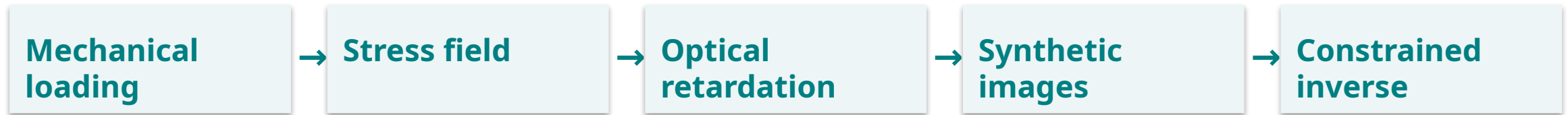


From mechanical stress to polarized-light images

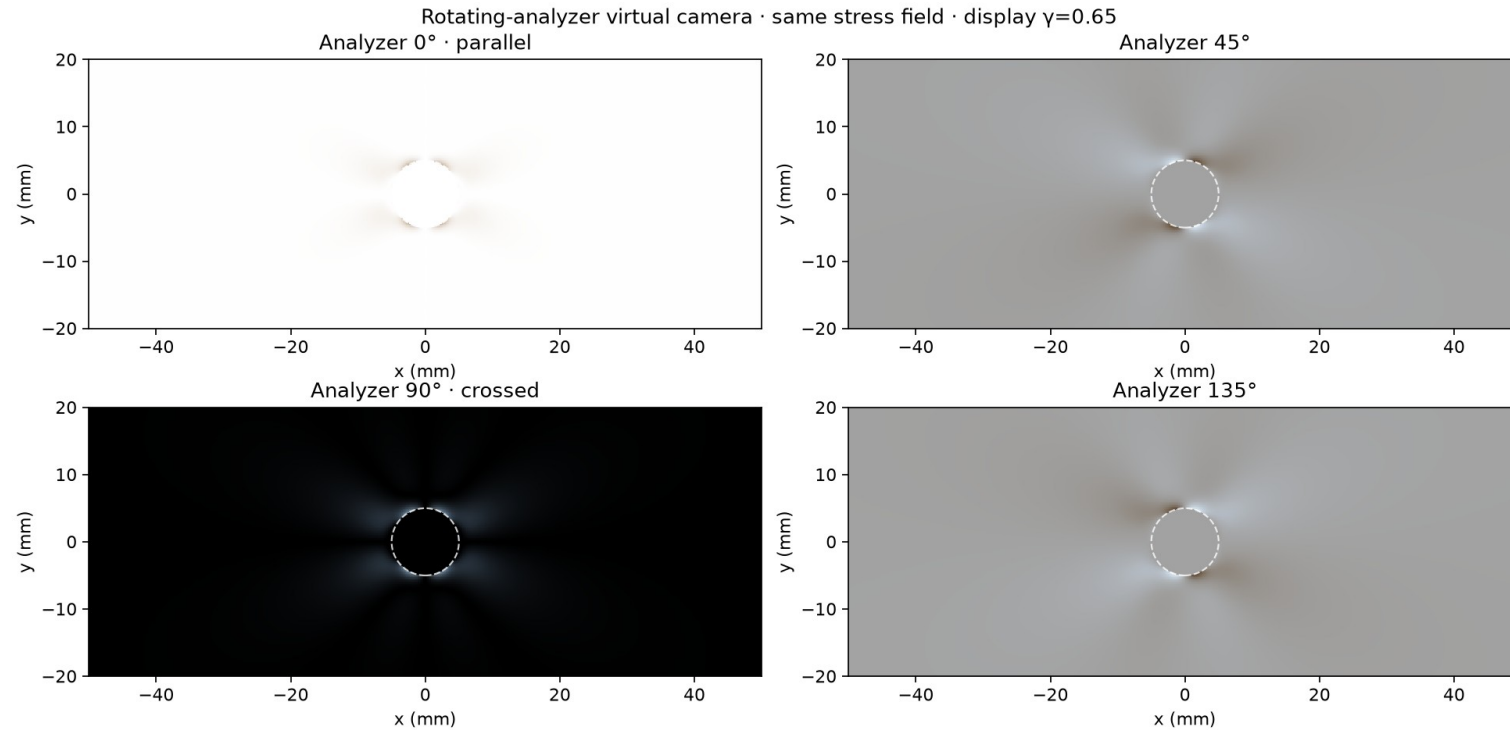
Objective

Predict the optical response around a circular hole and explore where stress differences can be recovered from synthetic images.



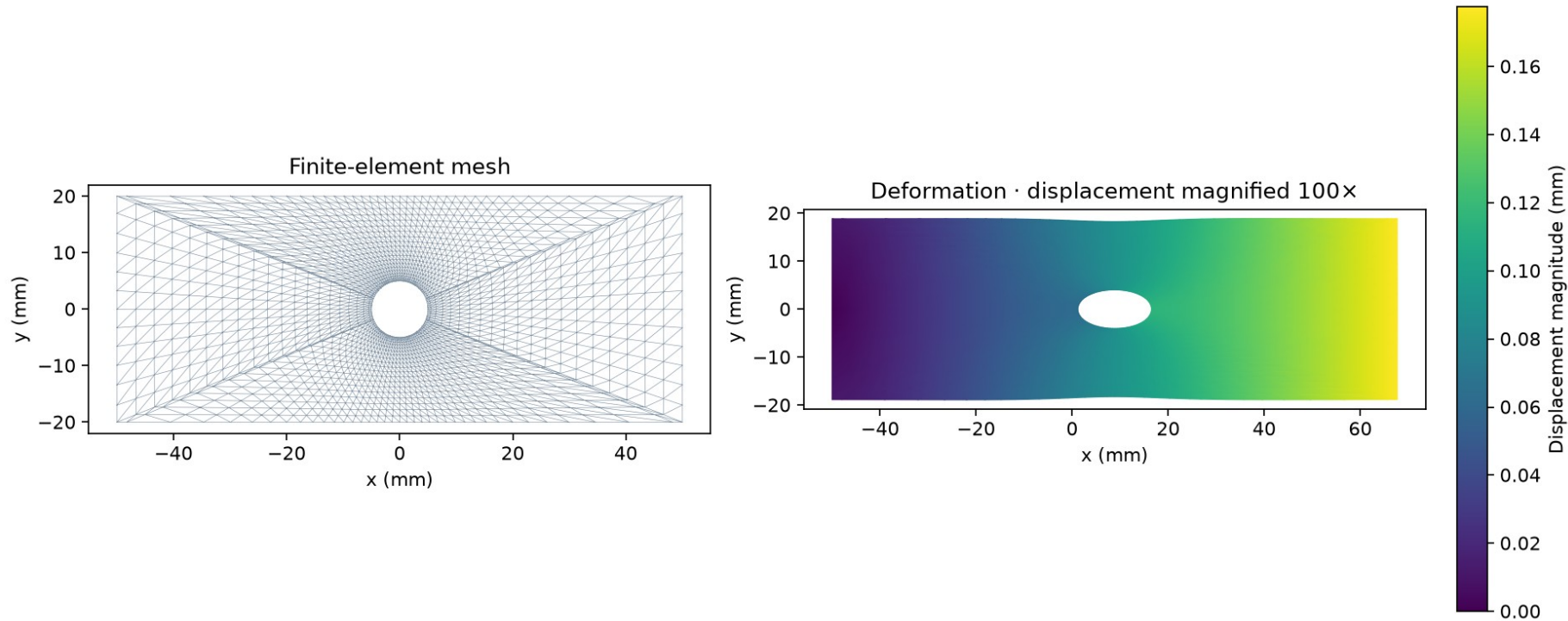
- Mechanics: analytical and finite-element models provide stress differences and principal-axis orientations.
- Optics: the stress-optic law and Jones model predict wavelength- and analyzer-dependent intensity.
- Inverse: recover stress differences only where measurements are informative, with explicit uncertainty.

Image 1 — Effect of analyzer rotation



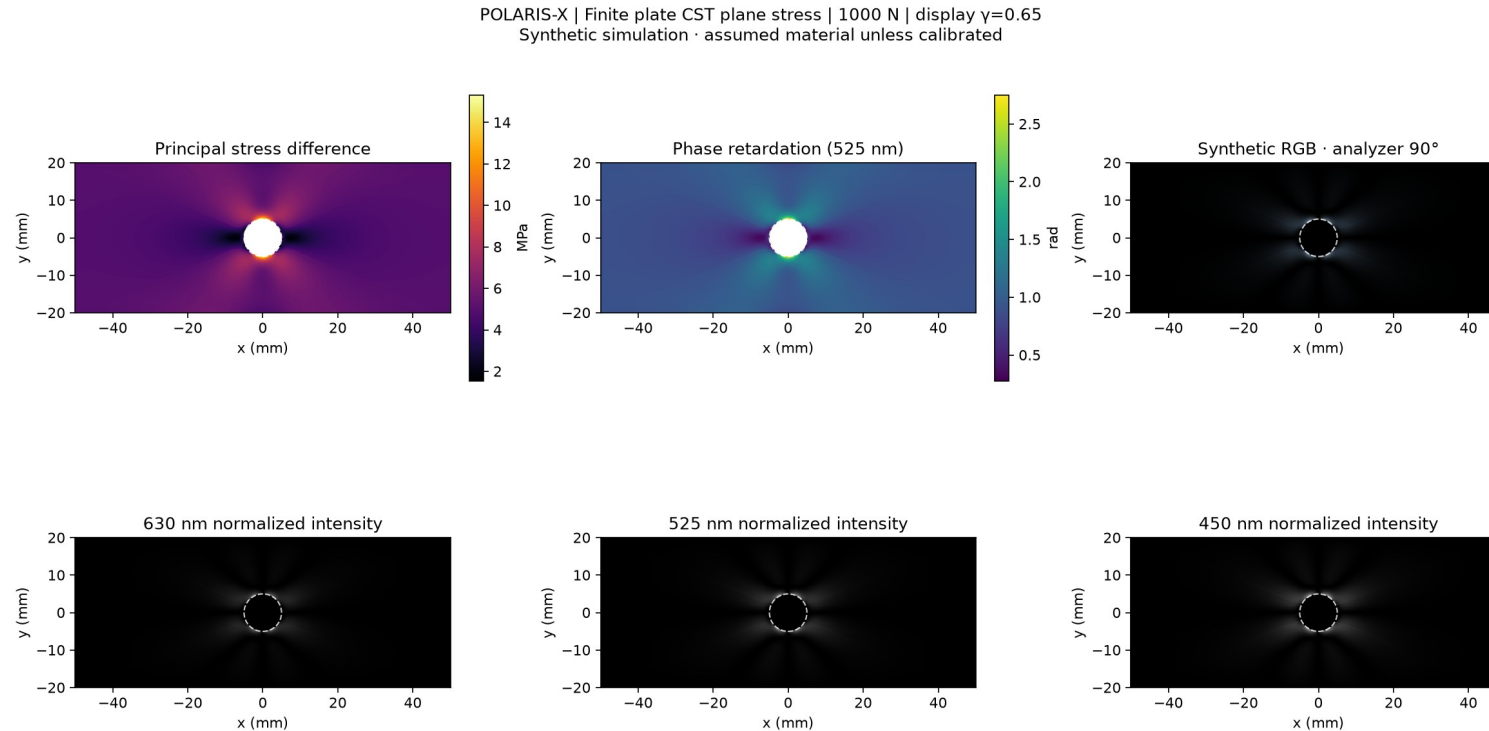
- Same Kirsch field at nominal 1000 N, viewed at analyzer angles of 0°, 45°, 90° and 135°.
- Parallel polarizers give a bright background; crossed polarizers reveal localized optical contrast.
- Image brightness depends on optical orientation and retardation, as well as the display gamma.

Image 2 — Finite-element mesh and deformation



- Finite plate at 1000 N: triangular elements are refined near the hole.
- The deformed shape is magnified 100x to make small displacements visible.
- The color scale gives actual displacement magnitude in mm; its values are not magnified.

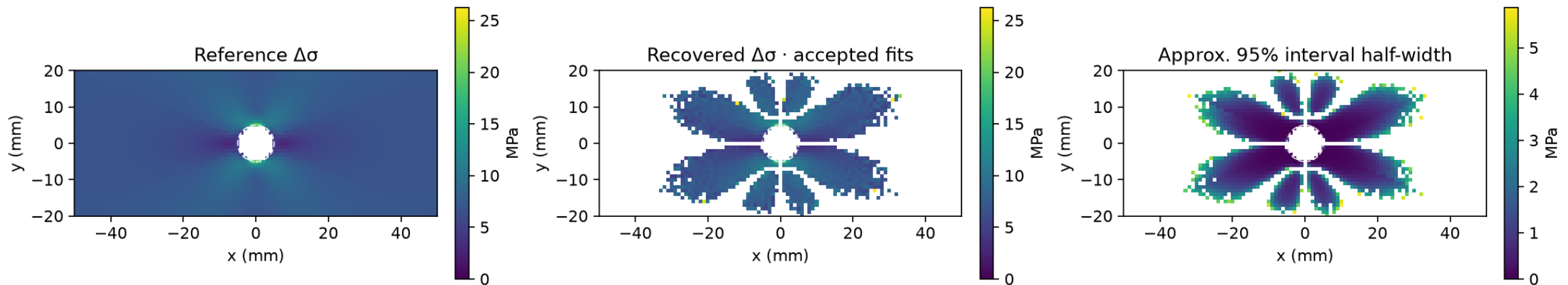
Image 3 — Finite-plate stress-to-light simulation



- Finite plate at 1000 N: principal stress difference is elevated near the hole boundary.
- The stress-optic law converts stress difference to retardation; Jones optics predicts the intensity.
- Synthetic RGB combines 630, 525 and 450 nm channels, shown separately below.

Image 4 — Constrained reconstruction at 1400 N

Synthetic constrained inversion · 1400 N · known principal-axis orientation
Accepted material pixels: 34.3% · RMSE 1.42 MPa · local intervals exclude calibration/model error



- Known orientation is supplied; 34.3% of material pixels pass the acceptance checks.
- RMSE: 1.42 MPa; maximum error: 19.31 MPa. Acceptance does not guarantee correctness.
- Blank regions are unresolved or outside material. Local intervals exclude model/calibration error.